

```

- name: "MYSQL_DATABASE"
  value: "glarimy"
ports:
- containerPort: 3306
  name: mysqldb
volumeMounts:
- name: mysql-persistent-storage
  mountPath: /var/lib/mysql
volumes:
- name: mysql-persistent-storage
  persistentVolumeClaim:
    claimName: mysqldb
---
apiVersion: v1
kind: PersistentVolume
metadata:
  name: mysqldb
  labels:
    type: local
spec:
  storageClassName: manual
  capacity:
    storage: 20Gi
  accessModes:
    - ReadWriteOnce
  hostPath:
    path: "/mnt/data"
---
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: mysqldb
spec:
  storageClassName: manual
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 2Gi

```

Deploying Kafka on Kubernetes

Apache Kafka cluster requires Zookeeper for internal management. So we need to deploy both. Since Kafka and Zookeeper have their own discovery protocol, we expose them on NodePort and connect them.

```

apiVersion: v1
kind: Service
metadata:
  labels:
    app: zookeeper

```

```

  name: zookeeper
spec:
  type: NodePort
  ports:
    - name: zookeeper-port
      port: 2181
      nodePort: 30181
      targetPort: 2181
  selector:
    app: zookeeper
---
apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app: zookeeper
    name: zookeeper
spec:
  selector:
    matchLabels:
      app: zookeeper
  replicas: 1
  template:
    metadata:
      labels:
        app: zookeeper
    spec:
      containers:
        - image: bitnami/zookeeper
          name: zookeeper
          ports:
            - containerPort: 2181
          env:
            - name: ZOOKEEPER_ID
              value: "1"
            - name: ZOOKEEPER_SERVER_1
              value: zookeeper
            - name: ALLOW_ANONYMOUS_LOGIN
              value: "yes"
---
apiVersion: v1
kind: Service
metadata:
  labels:
    app: kafka
    name: kafka
spec:
  type: NodePort
  ports:
    - name: kafka-port
      port: 9092
      nodePort: 30092

```